Week 29: Two-length builders

Adult guide built from the problem files · Unpiloted. Physical pretests unperformed.

Each problem's key repeats the main answer from the print guide, week-29-facilitator.pdf, whose source is editable/facilitator-src/guide.json; the hints and the proof of the last-gap formula are on the guide page named in the key's first line. The overview repeats the guide's first page. Change both together; check_guide_copy.py checks that every sentence here is in the guide.

Mathematical overview

The mathematics in one view

With unlimited copies of two positive integer rod lengths a and b, a target n is reachable exactly when $n = ax + by$ for nonnegative integers x, y. Rod order does not matter; cutting, overlaps, and negative rods are forbidden.

Finite evidence can certify infinitely many targets. If a consecutive integers $N, \ldots, N + a - 1$ are reachable, then every integer at least N is reachable: repeatedly add an a-rod to these a starting builds.

Exact boundary. If $\gcd(a, b) = 1$ and $a, b > 1$, the largest impossible target is $ab - a - b$, so every target at least $(a - 1)(b - 1)$ works. If $\gcd(a, b) > 1$, every target not divisible by that gcd is impossible, giving infinitely many gaps. A 1-rod makes every nonnegative target possible, so there is no largest positive gap in that case.

What each group is doing

K–1: Investigate small builds, distinguish rearranging from a new combination, and physically continue a run. Counting and alignment to 18 are enough; adults can record. A repeated construction is a genuine explanation without the formula.

Grades 2–3: Make exhaustive small lists, then find a short certificate for no later gaps. Addition to 28 and willingness to organize cases help. The shared-factor example refutes an overbroad conjecture.

Grades 4–5: Explain completeness, compare last gaps, conjecture the formula, and justify eventual coverage for a chosen coprime pair. Multiplication and remainders support the last questions; the general formula proof is adult backup.

Problems and routes

Problem	K–1	2–3	4–5	route
Which lengths 3-rods and 4-rods build	P1	P1	P1	first
Every way to build 12, 15, 16 and 24 with 3-rods and 4-rods	P2	P2		first
Last impossible lengths for 3 and 5, and for 4 and 5			P2	first
Which lengths 3-rods and 5-rods build	P3	P3		first
Lengths 18 through 29 with 4-rods and 7-rods			P3	return
Can 4-rods and 7-rods make 17?			P4	return
Every length from some point on	P4	P4		return
Lengths 1 through 12 with 2-rods and 4-rods	P5			return
The last impossible length for 4-rods and 7-rods		P5		return
Does every pair eventually have no more gaps?		P6		return
Which pairs from 2, 3, 4 and 5 build every target 6 through 12	P6			return
Which pairs have infinitely many impossible lengths			P5	return
All the ways to build 24, and the exchange between them			P6	return
Predict and test the last impossible length for 5 and 7			P7	return
Invented rods with no common divisor			P8	return

First route. K–1 P1–P3; Grades 2–3 P1–P3; Grades 4–5 P1–P2. **Return visits.** K–1 P4–P6; Grades 2–3 P4–P6; Grades 4–5 P3–P8.

Key, one entry per problem

Which lengths 3-rods and 4-rods build

K–1 P1 · Grades 2–3 P1 · Grades 4–5 P1 · first route · wording differs by band

Idea. With 3-rods and 4-rods exactly 1, 2 and 5 are impossible; 6, 7 and 8 plus added 3-rods reach every longer length.

K–1: *Print guide, p. 2.* Of the listed targets, 6=3+3, 8=4+4, 9=3+3+3, 10=3+3+4 work. The targets 1, 2, 5 do not. One rod is too long for 1 or 2; for 5, one rod is too short and two rods total at least 6.

Grades 2–3: *Print guide, p. 3.* In 1-16, only 1,2,5 fail. The certificate 6,7,8 plus added 3-rods covers all later targets.

Grades 4–5: *Print guide, p. 3.* Exactly 1,2,5. Check these small failures and use the three-build certificate above; a long tested list alone does not prove completeness.

Every way to build 12, 15, 16 and 24 with 3-rods and 4-rods

K–1 P2 · Grades 2–3 P2 · first route · wording differs by band

Idea. Fixing the number of 4-rods fixes the rest, and the count of 4-rods steps by 3: 12 and 15 have two ways each, 16 two, 24 three.

K–1: *Print guide, p. 2.* For 12: four 3-rods or three 4-rods. For 15: five 3-rods, or one 3-rod and three 4-rods. These are all: after choosing the number of 4-rods, the remainder must be a multiple of 3. The third strip under each target stays empty.

Grades 2–3: *Print guide, p. 3.* Writing (number of 3-rods, number of 4-rods): 12 has (4,0),(0,3); 16 has (4,1),(0,4);

24 has (8,0),(4,3),(0,6). The number of 4-rods has one required remainder modulo 3, so stepping by three 4-rods lists every possibility.

Last impossible lengths for 3 and 5, and for 4 and 5

Grades 4–5 P2 · first route

Idea. The last gaps are 7 for 3/5 and 11 for 4/5, each settled by a run of consecutive builds as long as the shorter rod.

Print guide, p. 3. Last gaps: 3/5 $\to$ 7; 4/5 $\to$ 11. Certificates: 8,9,10 as above; 12=4+4+4, 13=4+4+5, 14=4+5+5, 15=5+5+5. For gap 11 with 4/5, zero, one, two 5-rods leave 11,6,1, none divisible by 4.

Which lengths 3-rods and 5-rods build

K–1 P3 · Grades 2–3 P3 · first route · wording differs by band

Idea. With 3-rods and 5-rods exactly 1, 2, 4 and 7 are impossible; 8, 9 and 10 plus added 3-rods reach every longer length.

K–1: *Print guide, p. 2.* With 3 and 5, 4 and 7 are impossible. The others work: 8=3+5, 9=3+3+3, 10=5+5, 11=3+3+5, 12=3+3+3+3. For 7, no 5 leaves a multiple of 3: the possible remainders are 7 and 2.

Grades 2–3: *Print guide, p. 3.* The impossible lengths in 1-20 are 1,2,4,7. The last is 7: 8=3+5, 9=3+3+3, 10=5+5, followed by added 3-rods, settle all later targets.

Lengths 18 through 29 with 4-rods and 7-rods

Grades 4–5 P3 · return route

Idea. Four consecutive builds, 18 to 21, plus added 4-rods settle every target from 18 on.

Print guide, p. 3. All twelve targets 18-29 work: use the four builds 18=4+7+7, 19=4+4+4+7, 20=4+4+4+4+4, 21=7+7+7 and add 4 once or twice.

Can 4-rods and 7-rods make 17?

Grades 4–5 P4 · return route

Idea. 17 is impossible with 4 and 7: zero, one or two 7-rods leave 17, 10 or 3, none a multiple of 4.

Print guide, p. 3. Target 17 fails: zero, one or two 7-rods leave 17, 10 or 3, none divisible by 4.

Every length from some point on

K–1 P4 · Grades 2–3 P4 · return route · wording differs by band

Idea. Three consecutive builds with 3-rods (6, 7, 8 for 3/4; 8, 9, 10 for 3/5) grow by added 3-rods into every longer length.

K–1: *Print guide, p. 2.* Start with 6=3+3, 7=3+4, 8=4+4; add 3 to reach 9,10,11, then 12,13,14, then 15,16,17, then 18. These three chains keep going forever and cover every integer at least 6.

Grades 2–3: *Print guide, p. 3.* Yes to both. Minimal consecutive certificates are 6,7,8 for 3/4 and 8,9,10 for 3/5: add 3-rods to each, so the gaps stop at 5 and 7. Any other finite collection that demonstrably covers every remainder and starts beyond the last gap is acceptable, such as 17,18,19 or 21,22,23.

Lengths 1 through 12 with 2-rods and 4-rods

K–1 P5 · return route

Idea. 2-rods and 4-rods build exactly the even lengths, since both are made of whole pairs of units.

Print guide, p. 2. Exactly 2,4,6,8,10,12 work. Both rods are made of whole pairs of units; joining them always leaves an even length. Every positive even length is made from 2-rods.

The last impossible length for 4-rods and 7-rods

Grades 2–3 P5 · return route

Idea. The last gap for 4/7 is 17; 18, 19, 20 and 21 plus added 4-rods settle every longer length.

Print guide, p. 3. Last gap 17. A certificate is 18=4+7+7, 19=4+4+4+7, 20=4+4+4+4+4, 21=7+7+7; then add 4-rods. For 17, using zero, one or two 7-rods leaves 17,10 or 3; none is divisible by 4.

Does every pair eventually have no more gaps?

Grades 2–3 P6 · return route

Idea. The claim is false: of the pairs from 2, 3, 4, 5 and 7, only 2 and 4 has gaps forever (every odd length); one counterexample refutes it.

Print guide, p. 3. The claim is false: 2/4 misses every odd length. Of the ten pairs from 2, 3, 4, 5 and 7 it is the only one with gaps forever; the other nine share no divisor larger than 1. The pair 4/7 does eventually have no gaps, as P5 proves. One counterexample is enough to refute “every pair.”

Which pairs from 2, 3, 4 and 5 build every target 6 through 12

K–1 P6 · return route

Idea. Of the six pairs, only 2 and 3, 2 and 5, and 3 and 4 build every target from 6 through 12.

Print guide, p. 2. Exactly {2,3}, {2,5}, {3,4} cover every target 6-12. For {2,3}, use 2-rods for even targets and one 3 plus 2-rods for odd targets. For {2,5}, do the same with one 5 for odd targets. The failures are {2,4}: 7,9,11; {3,5}: 7; {4,5}: 6,7,11. This checks all six distinct pairs.

Which pairs have infinitely many impossible lengths

Grades 4–5 P5 · return route

Idea. A pair has infinitely many gaps exactly when its lengths share a divisor larger than 1.

Print guide, p. 3. 2/4, 3/6 and 4/6 have infinitely many gaps (odd targets; nonmultiples of 3; odd targets and 2). The pairs 3/5 and 4/7 have only finitely many. The dividing line is a common divisor greater than 1; the coprime direction needs the argument on the next page.

All the ways to build 24, and the exchange between them

Grades 4–5 P6 · return route

Idea. 24 has three builds, (8,0), (4,3) and (0,6); any two differ by trading four 3-rods for three 4-rods.

Print guide, p. 3. All ways for 24 are (8,0),(4,3),(0,6). Replace four 3-rods by three 4-rods, or reverse that replacement. Equality of two builds gives $3\Delta x = -4\Delta y$, forcing changes to be multiples of this exchange.

Predict and test the last impossible length for 5 and 7

Grades 4–5 P7 · return route

Idea. The last gap for 5/7 is 23; with 5, 7, 11 and 17 from earlier pairs it fits ab - a - b.

Print guide, p. 4. Last gap for 5/7 is 23. It is impossible because zero to three 7-rods leave 23,16,9,2, none divisible by 5. A certificate is 24=5+5+7+7, 25=5×5, 26=5+7+7+7, 27=5+5+5+5+7, 28=7×4. Adding 5 covers all later targets. The earlier last gaps 5,7,11,17 and this 23 fit ab-a-b. These examples support a conjecture, not a proof of the formula.

Invented rods with no common divisor

Grades 4–5 P8 · return route

Idea. For coprime a and b, the multiples 0, b, ..., (a-1)b reach every remainder modulo a, so adding a-rods covers every long enough target.

Print guide, p. 4. Many choices work. Example: 2/5 has builds 4 and 5, so all targets at least 4 work. In general, assume a,b are positive and coprime. The a numbers 0,b,2b,...,(a-1)b have different remainders modulo a: equal remainders would make a divide (j-k)b and hence j-k, impossible for distinct indices in 0,...,a-1. Each remainder therefore has a reachable starting value; add a-rods within each remainder class. This proves eventual coverage for any invented pair. If a=1, the conclusion is immediate.